Sensor Metadata and Axis-Aware Tokenization Shape Zero-Shot Reconstruction in OPM-MEG

Anonymous submission

Abstract

Electrophysiological foundation models must accommodate changes in sensor hardware, including wearable optically pumped magnetometers (OPMs). We test how one pretrained model, BrainOmni, behaves when the recording hardware changes, using only input interventions and no weight updates. Relabelling radial OPM magnetometers as gradiometers raises zero-shot waveform reconstruction from 0.2501 to 0.3819 Pearson correlation in all 21 participants of a single-axis cohort, and moves the same way in a single-participant SQUID diagnostic. The effect is configuration-dependent: on a triaxial cohort, the monolithic tag contrast is $+0.0002$ ($p = 0.969$). A factorial crossing tag, filtering and partition shows that axis grouping beats an exactly size-matched random partition averaged over ten seeds under both tags. With soft homogeneous field correction (Soft-HFC), the advantage is $+0.1071$ under MAG and $+0.0745$ under GRAD, each in $10/10$ participants. The paired interaction is -0.0326: the grouping advantage is smaller under the GRAD tag. Reconstruction gains do not establish decoding gains. On tactile Braille classification, a classical decoder over six log band powers and four temporal window means per channel clears chance under Soft-HFC while the pretrained probe does not. At ten participants, the decoding results remain inconclusive. These findings motivate testing continuous sensor parameterization and axis-conditioned queries, with reconstruction and decoding evaluated separately against competitive baselines.

Introduction

LaBraM learns reusable EEG representations through channel patches and masked neural-code prediction (Jiang, Zhao, and Lu 2024). BrainOmni learns joint EEG and MEG representations through sensor-conditioned tokenization and masked-token modeling (Xiao et al. 2025). These approaches motivate testing sensitivity to recording hardware. Unlike fixed cryogenic SQUID arrays, OPMs permit flexible near-scalp placement; triaxial sensors measure three orthogonal field components (Rea et al. 2022).

OPM-MEG is therefore a stress test for the sensor-conditioning interface, which is where hardware enters the computation. We hold the weights fixed and intervene only on what the model is told about its sensors: the categorical tag, the position and orientation vectors, and the grouping of channels passed through spatial attention. The question is not whether the model can be fine-tuned onto OPM data, but which inputs its representation is sensitive to, and whether those changes survive as decoding gains.

Our contributions are: (i) a paired, per-participant quantification of sensor-type and geometry sensitivity on two public OPM cohorts with a SQUID control; (ii) a preprocessing-matched comparison of axis grouping against a random control with matching group sizes and pass count; and (iii) improved reconstruction without a detectable improvement in linear decoding against a competitive classical baseline.

Data and Evaluation

Datasets

Single-axis OPM. FACE-DEC, OpenNeuro ds005107 (Xu et al. 2024). We use 21 participants and the 50 channels matched by rounded position and orientation metadata. The documentation states that coordinates were approximated from an Elekta layout for display and are not measured OPM positions (OpenNeuroDatasets n.d.), so a shared channel set is not evidence of anatomical alignment and spatial perturbations test the model's response to supplied metadata.

Triaxial OPM. The 10 adult participants of Zenodo record 11126593 (Rier et al. 2024), 84 cached trials each, 155–176 retained channels, geometry in head coordinates. The task contrasts index-finger and little-finger stimulation; results apply to this adult subset, not the full developmental cohort.

SQUID control. Participant sub-01, run-01 of OpenNeuro ds000117 (Wakeman and Henson 2015), with 102 magnetometers and 204 planar gradiometers. This is a single-participant diagnostic used only to test whether the sensor-type effect is OPM-specific.

Model, Endpoint and Interventions

We evaluate the pretrained BrainOmni tiny configuration. Its tokenizer combines a temporal convolutional encoder with spatial cross-attention between 16 learned queries and per-channel embeddings of position, unit orientation and discrete sensor type, followed by residual vector quantization. The

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sequence backbone is bidirectional and trained by masked-token prediction over codebook entries, not autoregressively. Reconstruction uses the tokenizer encoder, quantizer and decoder with weights frozen.

The primary endpoint is Pearson correlation (PCC) between the reconstructed and the normalized input waveform window. Higher PCC means closer agreement with the observed input, which still contains artifact, not better source localization or denoising. The triaxial factorial averages per-trial PCC within participant.

Interventions on the single-axis array reassign the sensor tag from magnetometer (`MAG=1`) to gradiometer (`GRAD=2`), zero the orientation vectors, reverse them, or jointly permute position and orientation across channels, in every case leaving the waveforms untouched. All single-axis conditions share channel selection and a soft variant of homogeneous field correction (Tierney et al. 2021): Soft-HFC, $P = I - \alpha Q_R Q_R^\top$ with $\alpha = 0.25$, where Q_R is an orthonormal basis for the sensor-orientation subspace. Soft-HFC attenuates any component projecting onto that subspace, including neural signal, so its effect is separated from tokenization throughout; the supplement derives it. Conventional practice applies the full projector, as implemented in MNE-Python (MNE-Python Developers n.d.) and extended there to higher-order harmonic bases (Tierney et al. 2022); Seymour et al. (2022) review interference suppression for OPM arrays.

For triaxial arrays the monolithic tokenizer passes all retained channels through one spatial bank. Axis partitioning passes the X, Y and Z groups separately through the same pretrained tokenizer and reassembles the reconstructions, giving three 16-query passes instead of one. The random control uses ten seeds and exactly matches each participant's Z, X and Y group sizes. We average seeds within participant before inference. Crossing MAG/GRAD tags, raw/Soft-HFC inputs and monolithic/random/axis partitions gives a $2\times2\times3$ factorial; contrasts hold filtering constant.

Statistics

All comparisons are paired within participant. We report the mean paired difference with a percentile bootstrap confidence interval over participants (20,000 resamples), the number of participants in which the difference has the stated sign, and an exact two-sided sign-flip permutation p-value enumerating all 2^n assignments. With $n = 10$ the smallest attainable p is $2/2^{10} \approx 0.002$, so triaxial p-values at that value indicate perfect consistency rather than a precisely estimated tail. Tests are exploratory and unadjusted for multiple comparisons. The supplement gives participant values, seed sensitivity and the complete decoding comparisons.

Results

A Discrete Label Dominates Reconstruction

Table 1 and Figure 1A,B give the single-axis results; per-participant values are in supplement Tables S1 and S2. Relabelling the array from magnetometer to gradiometer, with waveforms, positions and orientations unchanged, raises reconstruction from 0.2501 to 0.3819, a paired difference of $+0.1318$ (95% CI $[+0.1275, +0.1360]$), a relative increase of 52.7%. The effect holds in 21 of 21 participants with no overlap between the two distributions, exact $p < 10^{-5}$. The SQUID control moves in the same direction and further: magnetometer-only reconstruction rises from 0.4105 to 0.7130 when those 102 native SQUID magnetometers are assigned the gradiometer category, a 73.7% relative increase, on hardware the model was pretrained on.

Condition	PCC [95% CI]	Δ (consist.)
Default (`MAG=1`)	.2501 [.2450, .2549]	baseline
Gradiometer tag	.3819 [.3779, .3859]	$+.1318$ (21/21)
Zero orientation	.2359 [.2295, .2421]	$-.0142$ (21/21)
Reversed orient.	.2340 [.2274, .2402]	$-.0161$ (21/21)
Scrambled geometry	.2330 [.2307, .2354]	$-.0171$ (19/21)

Table 1: Waveform reconstruction on FACE-DEC (21 participants, 50 shared channels, Soft-HFC $\alpha = 0.25$). Δ is the mean paired difference against the default tag; consistency counts participants moving in that direction. Coordinates are approximate, which limits anatomical interpretation of the geometry rows.

Because the intervention changes one categorical token and nothing numerical, the result isolates a property of the learned embedding rather than of the recording. It is consistent with the gradiometer subspace having absorbed most of the MEG pretraining signal. A simpler account is that the two embeddings carry learned gain and unit priors and that OPM amplitude statistics sit nearer the gradiometer prior. These predict differently under a continuous parameterization, and this design cannot separate them. The triaxial factorial below limits either account: a tag advantage on one cohort does not establish a general preference for the gradiometer category.

The geometry perturbations move in the opposite direction and are small: zeroing orientations costs -0.0142 ($21/21$ participants declined), reversing them -0.0161 ($21/21$ declined), and scrambling position and orientation jointly -0.0171 ($19/21$ declined). The tokenizer does read the directional vectors, but on this cohort the discrete tag is seven to nine times more consequential than the continuous geometry, depending on which perturbation is the comparator. Scrambling approximate coordinates may scramble little, so this bounds the tag against the metadata quality of this cohort rather than against geometry as such.

Tag and Axis Grouping Interact

Table 2 crosses tag, filtering and partition. Against the ten-seed random mean, axis partitioning gains $+0.1071$ ($[+0.1002, +0.1143]$) under MAG with Soft-HFC and $+0.1120$ ($[+0.1048, +0.1208]$) without it. Under GRAD the gains remain $+0.0745$ ($[+0.0656, +0.0833]$) and $+0.0688$ ($[+0.0567, +0.0805]$). All four contrasts are positive in $10/10$ participants ($p = 0.002$). Under Soft-HFC/MAG, the seed-specific cohort gains range from 0.0969 to 0.1178 (SD 0.0049), so the result persists across partitions.

The tag effect depends on partition. GRAD minus MAG is $+0.0002$ ($[-0.0104, +0.0106]$, $5/10$, $p = 0.969$) on

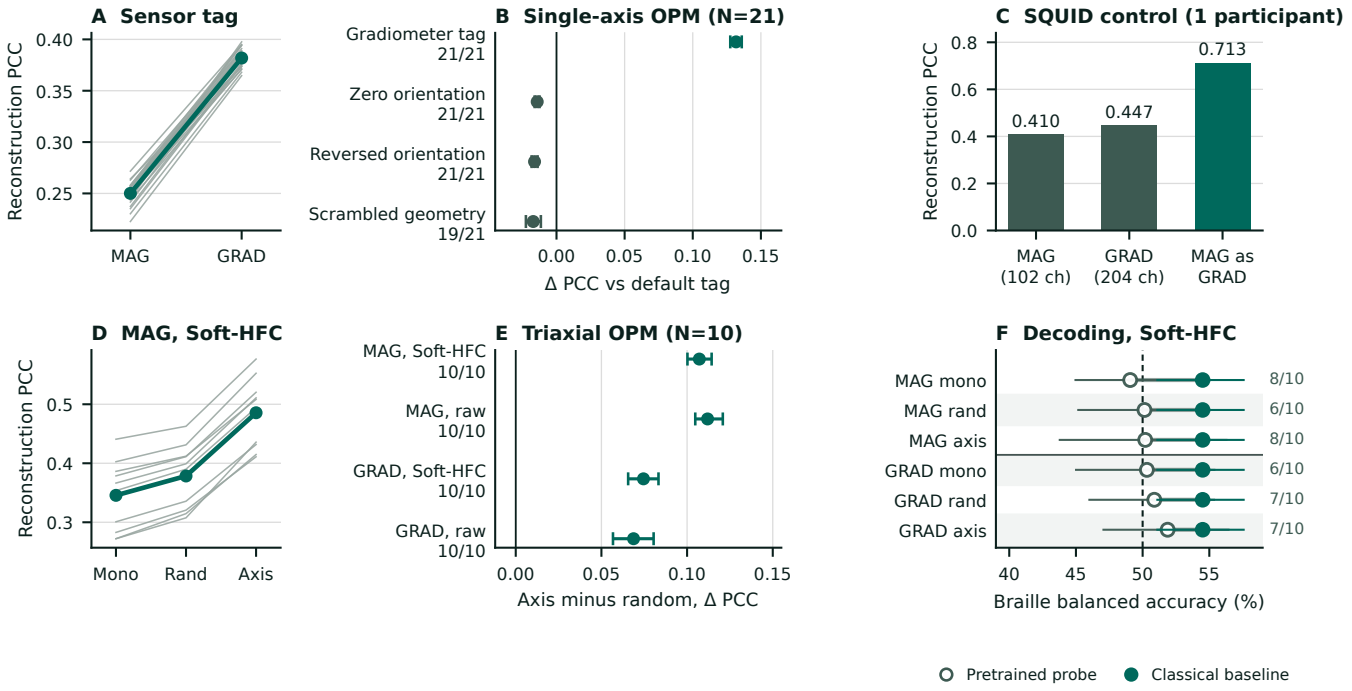


Figure 1: Paired per-participant reconstruction and decoding results. **A**, **D**: paired trajectories, one grey line per participant, cohort mean in colour; **D** traces MAG-tag monolithic, ten-seed random and axis partitions under Soft-HFC. **B**, **E**: mean paired differences with participant bootstrap 95% intervals; the count under each label gives participants moving in the stated direction. **B** is against the default magnetometer tag; **E** gives axis-minus-random contrasts for both tags and filters. **C**: single-participant SQUID diagnostic. **F**: full-array Braille decoding under Soft-HFC for both tags, probe (open) against classical baseline (filled), dashed line at chance; the right column counts participants where classical exceeds probe. Random-partition decoding uses seed 42 with axis-matched group sizes. The classical arm is invariant to tag and partition.

Partition	HFC	MAG PCC	GRAD PCC
Monolithic	no	0.3726	0.3728
Random	no	0.4019	0.4174
Axis	no	0.5138	0.4862
Monolithic	Soft	0.3456	0.3471
Random	Soft	0.3785	0.3978
Axis	Soft	0.4856	0.4723

Table 2: Full-array triaxial factorial, 10 adults, 84 trials each. Trial-weighted participant PCC; random partitions average ten seeds with group sizes matched to each participant's axes. Supplement Tables S3–S6 give intervals, paired contrasts, participant values and seed sensitivity.

raw monolithic inputs, and $+0.0014$ ($p = 0.824$) under Soft-HFC. These estimates give no evidence of the large FACE-DEC advantage, without establishing equivalence. Under Soft-HFC, remapping helps the random partition by $+0.0193$ ($p = 0.012$); on raw axis partitions it changes PCC by -0.0277 ($p = 0.006$). The paired change in axis-minus-random gain from MAG to GRAD is -0.0326 ($[-0.0431, -0.0233]$) under Soft-HFC and -0.0432 ($[-0.0548, -0.0329]$) raw, both $p = 0.002$. Grouping survives under both tags, but its advantage is smaller under GRAD, so the two effects are not additive on this scale.

The monolithic-to-random contrasts are tag-conditional too. Under Soft-HFC, random minus monolithic is $+0.0328$ under MAG and $+0.0507$ under GRAD. These account for 23.5% and 40.5% of the respective axis-over-monolithic gains. These are descriptive partition components: random partitioning changes attention composition as well as query budget. A matched-query monolithic control remains missing, so a capacity contribution cannot be isolated.

No Detectable Improvement in Linear Decoding

The full-array factorial evaluates tactile Braille decoding across both tags, both filtering regimes and all three partitions. Random-partition decoding uses seed 42 with axis-matched group sizes. Figure 1F shows the six Soft-HFC configurations; supplement Table S7 gives all twelve configurations and the random-initialization control. The pretrained linear probe has means from 47.53% to 51.88%, with every participant-bootstrap interval including chance. Paired axis-minus-monolithic and axis-minus-random decoding contrasts also have intervals spanning zero in every tag/filter combination (supplement Table S8).

The classical baseline uses six log band powers spanning 1–96 Hz plus four temporal window means per channel, classified by L_2-regularized logistic regression. It reaches 54.50% $[51.03, 57.65]$ under Soft-HFC and 53.54%

$[49.93, 56.75]$ without filtering. Its features depend on filtering but not on the sensor tag or tokenizer partition, so the six rows within each filtering regime share one classical measurement. Only the Soft-HFC interval excludes chance. The classical cohort mean exceeds the pretrained probe in all twelve configurations, although this ordering alone does not establish a significant paired advantage.

We observe improved reconstruction without a detectable improvement in linear decoding. With ten participants, these results remain inconclusive and do not establish equivalence. A larger cohort is needed to distinguish limitations of the representation from task difficulty.

Discussion and Limitations

Two design hypotheses follow. First, continuous sensor metadata merits testing against the categorical interface. A continuous parameterization, such as stand-off distance and baseline length, would let a new sensor occupy its own place instead of borrowing a category, and is testable against the categorical baseline. The 52.7% FACE-DEC gain is configuration-dependent; the triaxial monolithic estimates do not support its generalization, and the cohorts differ in more than axis count.

Second, axis-conditioned spatial queries merit testing in multi-axis arrays. Grouping retains an advantage under both tags, although its magnitude depends on the tag. The partition we tested is not free, though: it retains every channel but severs attention between the three orthogonal channels of each triaxial sensor, and it triples tokenizer compute. Axis-conditioned queries inside a single bank would test the same idea at neither cost, and would say whether cross-axis attention is harmful or merely unused.

The limits are real. We evaluate one model on two small OPM cohorts with a single-participant SQUID diagnostic, so no harmonized cross-hardware estimate follows. FACE-DEC coordinates are approximate. Ten random seeds and the crossed tag design strengthen the grouping result, but the monolithic tokenizer still lacks a matched token budget. The algebraic decomposition cannot identify capacity and grouping mechanisms separately. Every positive result uses correlation to the processed input, so a model can gain by matching artifact structure. An artifact-robust endpoint is needed to test whether the reconstruction gains reflect neural signal. Results are based on cached inputs and saved model outputs; raw-data replay remains untested.

Conclusion

Sensor metadata and channel grouping shape zero-shot reconstruction in configuration-dependent ways. The large single-axis tag effect is absent from the triaxial monolithic point estimates, while axis grouping retains an advantage under both tags. Their interaction shows that the grouping advantage depends on the sensor tag in this factorial. These reconstruction improvements do not establish better decoding; continuous sensor parameterization and axis-conditioned queries remain hypotheses to test against competitive baselines.

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Supplementary Material: Sensor Metadata and Axis-Aware Tokenization Shape Zero-Shot Reconstruction in OPM-MEG

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This supplement presents additional methods, participant-level reconstruction results, paired comparisons, decoding results, and dataset details.

S1 Soft Homogeneous Field Correction

S1.1 Setup

Following Tierney et al. (2021), we approximate distant magnetic interference as a spatially uniform field $\mathbf{B}_{\text{ext}} \in \mathbb{R}^3$. A sensor c with unit sensitive orientation $\mathbf{u}_c$ measures the projection $m_c = \mathbf{u}_c^\top \mathbf{B}_{\text{ext}}$, so across an array of C channels the induced measurement is $\mathbf{m}_{\text{ext}} = \mathbf{Q}\mathbf{B}_{\text{ext}}$, where row c of $\mathbf{Q} \in \mathbb{R}^{C\times 3}$ is $\mathbf{u}_c^\top$.

Conventional homogeneous field correction (Tierney et al. 2021) projects onto the null space of $\mathbf{Q}^\top$:

$$\mathbf{P}_{\text{hard}} = \mathbf{I}_C - \mathbf{Q}(\mathbf{Q}^\top\mathbf{Q})^{-1}\mathbf{Q}^\top.$$

When $\mathbf{Q}$ has full column rank, $\operatorname{rank}(\mathbf{P}_{\text{hard}}) = C-3$, making the projected channel covariance rank-deficient.

The reference implementation of this correction is `mne.preprocessing.compute_proj_hfc` in MNE-Python (MNE-Python Developers n.d.), whose `order` argument selects the homogeneous field alone or a higher-order harmonic basis (Tierney et al. 2022). Seymour et al. (2022) survey the interference-suppression options available for OPM arrays. We did not use that implementation: the variant evaluated here applies partial rather than full projection and is defined next.

S1.2 Soft variant

The cited method uses full projection. Here we evaluate a soft shrinkage variant parameterized by $\alpha \in [0, 1)$:

$$\mathbf{P}_{\text{soft}}(\alpha) = \mathbf{I}_C - \alpha\,\mathbf{Q}_R\mathbf{Q}_R^\top,$$

where $\mathbf{Q}_R$ is an orthonormal basis of $\operatorname{col}(\mathbf{Q})$, of dimension $r = \operatorname{rank}(\mathbf{Q}) \le 3$. Its eigenvalues are $1-\alpha$ on those r directions and 1 on the remaining $C-r$, so for any $\alpha < 1$ the operator has full rank C. The current implementation uses a rank-aware SVD to avoid extra QR directions for rank-deficient arrays. Historical results used thin QR and have not been recomputed after this correction.

The coefficient $\alpha = 0.25$ is an analysis choice here, not a value established by Tierney et al. (2021). At this value, components lying in the uniform-field subspace are scaled by 0.75 in amplitude, a power reduction of $1-0.75^2 = 43.75\%$.

S1.3 What this does to neural signal

A neural component orthogonal to $\operatorname{col}(\mathbf{Q})$ satisfies $\mathbf{Q}_R^\top\mathbf{s} = \mathbf{0}$ and is returned unchanged. This is not a claim that neural signal is preserved. A dipolar field pattern is not in general orthogonal to the uniform-field subspace, and whatever part of it does project onto $\operatorname{col}(\mathbf{Q})$ receives the same 25% amplitude shrinkage as the interference. Soft-HFC is therefore a lossy operation on signal as well as noise, which is why the main paper holds filtering constant within every contrast rather than treating it as a neutral preprocessing step.

S2 Single-Axis Metadata Interventions

All conditions share channel selection, trials and Soft-HFC. Only the metadata handed to the tokenizer differs, so any difference between columns of Table S1 is a property of the learned embedding rather than of the recording. Table S2 gives the paired tests.

The dataset documentation states that FACE-DEC sensor coordinates were approximated from an Elekta layout for topographic display and are not measured OPM positions. The geometry rows should therefore be read as sensitivity to supplied metadata, not as evidence about forward-field accuracy.

S3 Tag by Partition Factorial

The experiment crosses MAG/GRAD tags, raw/Soft-HFC inputs and monolithic/random/axis partitions. Random group sizes match each participant's Z, X and Y counts exactly in the generating script, with seeds 42 and 101–109. Saved group assignments are unavailable for an independent metadata-level check. We average seeds within participant before paired inference, retaining ten independent participants. Tables S3–S6 report the full-array reconstruction results.

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PCC is averaged over channels and temporal windows within each trial, then over trials within participant. Each participant contributes equally to cohort estimates.

The tag effect is configuration-dependent. The raw monolithic contrast is $+0.0002$ ($[-0.0104, +0.0106]$, $p = 0.969$), while axis grouping retains its advantage under both tags. The paired change in the grouping advantage under GRAD is -0.0432 raw and -0.0326 with Soft-HFC, both negative in every participant. This interaction does not by itself identify a shared mechanism. Under Soft-HFC, the monolithic-to-random contrast accounts for 23.5% under MAG and 40.5% under GRAD of the respective axis-over-monolithic gains. These are descriptive partition components: random partitioning changes attention composition as well as query budget, so a capacity contribution cannot be isolated. Tests are exploratory and unadjusted for multiple comparisons.

S4 Decoding

The classical decoder uses scikit-learn `LogisticRegression` with L_2 penalty at the library default $C = 1.0$, fixed for every configuration and never tuned on a held-out fold, so no selection leakage is possible. The pretrained probe uses the same solver and folds. The sensor-type codes are checkpoint-specific integers, `MAG=1` and `GRAD=2` in the released tiny configuration, and any reuse of this intervention must read them from the checkpoint rather than assume them.

Table S7 gives all twelve full-array configurations and three decoders, evaluated with five-fold cross-validation within participant. Random-partition decoding uses seed 42 with group sizes matched to the axes, while reconstruction averages ten seeds. Every pretrained-probe interval includes chance. Table S8 gives the paired axis-minus-monolithic and axis-minus-random probe contrasts; every interval includes zero. These are exploratory tests without multiplicity correction, not evidence of equivalence.

The classical features are six log band powers spanning 1–96 Hz plus four temporal window means per channel. They depend on filtering but not on sensor tag or tokenizer partition. The six classical rows within each filtering regime therefore share one underlying measurement. The Soft-HFC interval excludes chance, while the unfiltered interval does not. The classical cohort mean exceeds the pretrained probe in every configuration, but this ordering alone does not establish a significant paired advantage or rule out a task-level noise floor.

S5 Model Architecture

The pretrained BrainOmni tiny configuration is evaluated with frozen weights. The tokenizer conditions on sensor position, unit orientation, and a discrete sensor type, and uses residual vector quantization (Xiao et al. 2025). In the evaluated configuration, the sequence backbone is bidirectional (`causal=False`) and uses masked-token prediction over codebook entries with a mask ratio of 0.25. Attention operates within trials, not across trials. The number of learned spatial queries is loaded from the checkpoint.

Axis partitioning is an inference-time reorganization only. The X, Y and Z groups pass separately through the same frozen tokenizer and the reconstructions are reassembled, giving three spatial-attention passes instead of one. No weights change.

S6 Dataset Details

FACE-DEC, OpenNeuro `ds005107` v1.0.0, DOI `10.18112/openneuro.ds005107.v1.0.0`, 21 participants and the 50 channels matched across participants by rounded position and orientation metadata. Triaxial OPM, Zenodo record `10.5281/zenodo.11126593`, the 10 adult participants, 84 cached trials each, 155 to 176 retained channels. SQUID control, OpenNeuro `ds000117` v1.0.4, participant `sub-01`, run `01`.

Participant	Default	GRAD tag	Zero ori.	Rev. ori.	Scrambled
P01	0.2717	0.3950	0.2600	0.2563	0.2327
P02	0.2522	0.3735	0.2413	0.2418	0.2331
P03	0.2435	0.3713	0.2299	0.2268	0.2265
P04	0.2444	0.3976	0.2310	0.2250	0.2420
P05	0.2484	0.3944	0.2386	0.2445	0.2466
P06	0.2573	0.3910	0.2461	0.2439	0.2279
P07	0.2580	0.3940	0.2501	0.2511	0.2350
P08	0.2630	0.3890	0.2518	0.2502	0.2327
P09	0.2641	0.3736	0.2549	0.2504	0.2382
P10	0.2486	0.3708	0.2352	0.2306	0.2376
P11	0.2546	0.3855	0.2384	0.2384	0.2300
P12	0.2415	0.3839	0.2270	0.2241	0.2355
P13	0.2302	0.3680	0.2134	0.2063	0.2347
P14	0.2579	0.3864	0.2431	0.2425	0.2352
P15	0.2352	0.3753	0.2159	0.2109	0.2251
P16	0.2571	0.3836	0.2462	0.2446	0.2350
P17	0.2564	0.3884	0.2467	0.2425	0.2251
P18	0.2509	0.3745	0.2327	0.2307	0.2303
P19	0.2562	0.3813	0.2363	0.2370	0.2268
P20	0.2373	0.3771	0.2163	0.2137	0.2258
P21	0.2226	0.3647	0.1990	0.2020	0.2368
Mean	0.2501	0.3819	0.2359	0.2340	0.2330

Table S1: Zero-shot reconstruction PCC per participant on FACE-DEC (ds005107), 50 shared channels, Soft-HFC $\alpha = 0.25$. Participants are indexed in the saved record order, which is not the dataset's own participant numbering. Waveforms are identical across columns; only the metadata supplied to the tokenizer differs.

Contrast vs default	Mean Δ	95% CI	Consistent	Exact p
Gradiometer tag	$+0.1318$	$[+0.1275, +0.1360]$	21/21	9.54e-07
Zero orientation	-0.0142	$[-0.0160, -0.0124]$	21/21	9.54e-07
Reversed orientation	-0.0161	$[-0.0183, -0.0139]$	21/21	9.54e-07
Scrambled geometry	-0.0171	$[-0.0225, -0.0115]$	19/21	4.58e-05

Table S2: Paired single-axis contrasts, 21 participants. Confidence intervals are percentile bootstrap over participants, 20,000 resamples, seed 0. The consistency column counts participants moving in the stated direction. p is the exact two-sided sign-flip permutation value enumerating all 2^{21} assignments.

Filter	Tag	Partition	Mean PCC	95% CI
Raw	MAG	Monolithic	0.3726	[0.3409, 0.4047]
Raw	MAG	Random	0.4019	[0.3721, 0.4316]
Raw	MAG	Axis	0.5138	[0.4846, 0.5441]
Raw	GRAD	Monolithic	0.3728	[0.3388, 0.4064]
Raw	GRAD	Random	0.4174	[0.3890, 0.4454]
Raw	GRAD	Axis	0.4862	[0.4652, 0.5066]
Soft-HFC	MAG	Monolithic	0.3456	[0.3102, 0.3807]
Soft-HFC	MAG	Random	0.3785	[0.3457, 0.4105]
Soft-HFC	MAG	Axis	0.4856	[0.4512, 0.5201]
Soft-HFC	GRAD	Monolithic	0.3471	[0.3141, 0.3796]
Soft-HFC	GRAD	Random	0.3978	[0.3702, 0.4247]
Soft-HFC	GRAD	Axis	0.4723	[0.4500, 0.4944]

Table S3: Full-array factorial reconstruction. Trial-weighted participant PCC; random scores average ten seeds within participant. All intervals use 20,000 participant bootstrap draws with seed 0. Neither seeds nor trials are treated as independent participants.

Filter	Contrast	Mean Δ	95% CI	Positive	Exact p
Raw	MAG: Axis minus random	$+0.1120$	$[+0.1048, +0.1208]$	10/10	1.95e-03
Raw	MAG: Random minus mono	$+0.0292$	$[+0.0262, +0.0322]$	10/10	1.95e-03
Raw	GRAD: Axis minus random	$+0.0688$	$[+0.0567, +0.0805]$	10/10	1.95e-03
Raw	GRAD: Random minus mono	$+0.0446$	$[+0.0390, +0.0504]$	10/10	1.95e-03
Raw	GRAD minus MAG: Monolithic	$+0.0002$	$[-0.0104, +0.0106]$	5/10	9.69e-01
Raw	GRAD minus MAG: Random	$+0.0155$	$[+0.0065, +0.0234]$	9/10	1.37e-02
Raw	GRAD minus MAG: Axis	-0.0277	$[-0.0408, -0.0150]$	1/10	5.86e-03
Raw	Grouping interaction	-0.0432	$[-0.0548, -0.0329]$	0/10	1.95e-03
Soft-HFC	MAG: Axis minus random	$+0.1071$	$[+0.1002, +0.1143]$	10/10	1.95e-03
Soft-HFC	MAG: Random minus mono	$+0.0328$	$[+0.0292, +0.0363]$	10/10	1.95e-03
Soft-HFC	GRAD: Axis minus random	$+0.0745$	$[+0.0656, +0.0833]$	10/10	1.95e-03
Soft-HFC	GRAD: Random minus mono	$+0.0507$	$[+0.0448, +0.0564]$	10/10	1.95e-03
Soft-HFC	GRAD minus MAG: Monolithic	$+0.0014$	$[-0.0108, +0.0120]$	6/10	8.24e-01
Soft-HFC	GRAD minus MAG: Random	$+0.0193$	$[+0.0083, +0.0287]$	9/10	1.17e-02
Soft-HFC	GRAD minus MAG: Axis	-0.0133	$[-0.0276, +0.0008]$	5/10	1.15e-01
Soft-HFC	Grouping interaction	-0.0326	$[-0.0431, -0.0233]$	0/10	1.95e-03

Table S4: Paired factorial contrasts. Positive counts strictly positive participant differences, including when the mean is negative. The grouping interaction is $(A - R)_{\mathrm{GRAD}} - (A - R)_{\mathrm{MAG}}$, with R averaged across seeds within participant. Exact two-sided sign-flip tests are exploratory and unadjusted for multiplicity.

Filter	Participant	MAG mono	MAG rand	MAG axis	GRAD mono	GRAD rand	GRAD axis
Raw	P01	0.3183	0.3502	0.4505	0.3285	0.3788	0.4459
Raw	P02	0.3391	0.3675	0.4694	0.3162	0.3675	0.4499
Raw	P03	0.3060	0.3429	0.4520	0.3009	0.3597	0.4386
Raw	P04	0.3066	0.3370	0.4821	0.2989	0.3571	0.4610
Raw	P05	0.4338	0.4579	0.5771	0.4601	0.4896	0.5253
Raw	P06	0.4617	0.4824	0.5969	0.4315	0.4672	0.5320
Raw	P07	0.4046	0.4287	0.5250	0.4008	0.4425	0.4847
Raw	P08	0.3740	0.4076	0.5122	0.3962	0.4374	0.5175
Raw	P09	0.3856	0.4144	0.5302	0.3905	0.4323	0.5088
Raw	P10	0.3967	0.4302	0.5430	0.4046	0.4419	0.4981
Soft-HFC	P01	0.2828	0.3206	0.4110	0.3029	0.3602	0.4313
Soft-HFC	P02	0.3008	0.3357	0.4317	0.2894	0.3471	0.4333
Soft-HFC	P03	0.2722	0.3146	0.4148	0.2767	0.3416	0.4236
Soft-HFC	P04	0.2721	0.3075	0.4358	0.2782	0.3398	0.4403
Soft-HFC	P05	0.4026	0.4312	0.5526	0.4312	0.4651	0.5181
Soft-HFC	P06	0.4408	0.4627	0.5766	0.3997	0.4438	0.5266
Soft-HFC	P07	0.3863	0.4120	0.5082	0.3766	0.4240	0.4771
Soft-HFC	P08	0.3536	0.3900	0.4941	0.3677	0.4215	0.4977
Soft-HFC	P09	0.3664	0.3991	0.5108	0.3709	0.4143	0.4931
Soft-HFC	P10	0.3786	0.4112	0.5203	0.3776	0.4202	0.4819

Table S5: Full-array factorial PCC per participant. Indices follow sorted dataset participant IDs. Random columns average the ten saved seeds; other columns use one condition per participant.

Seed	Raw MAG	Raw GRAD	Soft-HFC MAG	Soft-HFC GRAD
42	+0.1124	+0.0663	+0.1083	+0.0745
101	+0.1110	+0.0706	+0.1062	+0.0768
102	+0.1114	+0.0713	+0.1057	+0.0768
103	+0.1115	+0.0636	+0.1066	+0.0683
104	+0.1147	+0.0684	+0.1085	+0.0742
105	+0.1215	+0.0781	+0.1178	+0.0836
106	+0.1136	+0.0725	+0.1083	+0.0781
107	+0.1009	+0.0591	+0.0969	+0.0638
108	+0.1140	+0.0677	+0.1092	+0.0739
109	+0.1087	+0.0699	+0.1040	+0.0754
SD	0.0049	0.0049	0.0049	0.0051

Table S6: Seed-specific cohort axis-minus-random PCC contrasts, trial-weighted. SD is the descriptive population SD across the ten observed seed-specific means (divisor ten), not a standard error. Group sizes exactly match axis counts in the generating script. Every participant contrast is positive for every seed in all four tag/filter combinations.

Filter	Tag	Partition	Probe [95% CI]	Ctrl A	Classical [95% CI]	Cl. > Pr.
Raw	MAG	Monolithic	48.36 [43.83, 52.68]	49.12	53.54 [49.93, 56.75]	8/10
Raw	MAG	Random	49.40 [45.00, 53.26]	50.74	53.54 [49.93, 56.75]	7/10
Raw	MAG	Axis	47.53 [43.18, 51.85]	49.53	53.54 [49.93, 56.75]	7/10
Raw	GRAD	Monolithic	48.89 [41.83, 54.68]	48.25	53.54 [49.93, 56.75]	6/10
Raw	GRAD	Random	51.79 [47.35, 55.36]	49.56	53.54 [49.93, 56.75]	6/10
Raw	GRAD	Axis	49.93 [45.15, 54.58]	48.83	53.54 [49.93, 56.75]	7/10
Soft-HFC	MAG	Monolithic	49.07 [44.89, 52.79]	49.69	54.50 [51.03, 57.65]	8/10
Soft-HFC	MAG	Random	50.14 [45.10, 54.54]	50.22	54.50 [51.03, 57.65]	6/10
Soft-HFC	MAG	Axis	50.19 [43.71, 56.36]	48.49	54.50 [51.03, 57.65]	8/10
Soft-HFC	GRAD	Monolithic	50.32 [44.93, 55.46]	49.25	54.50 [51.03, 57.65]	6/10
Soft-HFC	GRAD	Random	50.88 [45.93, 55.43]	51.21	54.50 [51.03, 57.65]	7/10
Soft-HFC	GRAD	Axis	51.88 [46.99, 56.51]	48.97	54.50 [51.03, 57.65]	7/10

Table S7: Full-array factorial Braille balanced accuracy (%), with participant bootstrap intervals. Ctrl A denotes the random-initialization control. Random decoding is available only at seed 42, with group sizes exactly matched to the axes; reconstruction averages ten seeds. Every pretrained-probe interval includes chance; this does not establish equivalence.

Filter	Tag	Probe contrast	Mean Δ	95% CI	Positive	Exact p
Raw	MAG	Axis minus mono	-0.8333	$[-4.1111, +2.3333]$	4/10	6.45e-01
Raw	MAG	Axis minus random	-1.8750	$[-5.3889, +1.2083]$	5/10	3.52e-01
Raw	GRAD	Axis minus mono	$+1.0417$	$[-3.1111, +5.3194]$	5/10	6.56e-01
Raw	GRAD	Axis minus random	-1.8611	$[-5.5000, +1.6389]$	4/10	3.57e-01
Soft-HFC	MAG	Axis minus mono	$+1.1250$	$[-1.5694, +4.1528]$	5/10	5.29e-01
Soft-HFC	MAG	Axis minus random	$+0.0556$	$[-4.4444, +4.7087]$	4/10	9.90e-01
Soft-HFC	GRAD	Axis minus mono	$+1.5556$	$[-1.0833, +4.2778]$	7/10	3.09e-01
Soft-HFC	GRAD	Axis minus random	$+1.0000$	$[-1.3333, +3.2917]$	6/10	4.45e-01

Table S8: Paired pretrained-probe decoding contrasts in balanced-accuracy percentage points, ten participants. Random partitions use seed 42. Intervals use 20,000 participant bootstrap draws; exact two-sided sign-flip tests are exploratory and unadjusted for multiplicity. Positive counts strictly positive participant differences. Every interval includes zero; this does not establish equivalence.

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